

Algebraic Theory of $\hat{L}$

$\hat{J}$ can be $\hat{L}$ or $\hat{S}$ or something else

- $[\hat{J}_i, \hat{J}_j] = i\hbar \epsilon_{ijk} \hat{J}_k$
- $[\hat{J}_i, \hat{J}^2] = 0$

$$\left. \begin{aligned} \hat{J}_+ &= \hat{J}_x + i\hat{J}_y \\ \hat{J}_- &= \hat{J}_x - i\hat{J}_y \end{aligned} \right\} (\hat{J}_+)^{\dagger} = \hat{J}_-$$

$$[\hat{J}_+, \hat{J}_-] = 2\hbar \hat{J}_z$$

$$\hat{J}_x^2 + \hat{J}_y^2 = \hat{J}_- \hat{J}_+ + \hbar \hat{J}_z$$

$$\Rightarrow \hat{J}^2 = \hat{J}_+ \hat{J}_- + \hat{J}_z^2 - \hbar \hat{J}_z = \hat{J}_- \hat{J}_+ + \hat{J}_z^2 + \hbar \hat{J}_z$$

- $[\hat{J}_{\pm}, \hat{J}^2] = 0$
- $[\hat{J}_z, \hat{J}_{\pm}] = \pm \hbar \hat{J}_{\pm}$

$\hat{J}_+$ adds one $\hbar$ to $\hat{J}_z$

$\hat{J}_-$ subtracts one $\hbar$ from $\hat{J}_z$

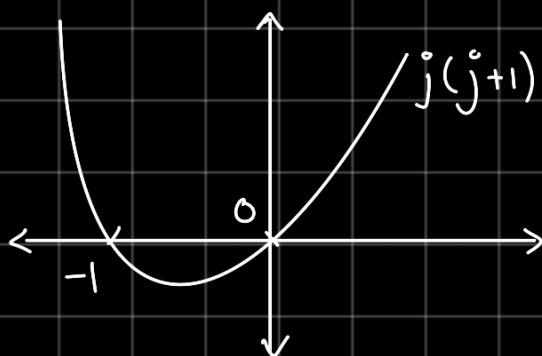
$\Rightarrow$ Orthonormality & Eigenstates

Eigenstate: $|j, m\rangle$: $j \rightarrow \hat{J}^2$ eigenvalue
 $m \rightarrow \hat{J}_z$ eigenvalue

$$\Rightarrow \hat{J}^2 |j, m\rangle = \hbar^2 j(j+1) |j, m\rangle$$

$$\hat{J}_z |j, m\rangle = \hbar m |j, m\rangle$$

note: $j(j+1) \geq 0 \rightarrow$ we can use both values of j
but we choose $j \geq 0$



Now for $\hat{J}_{\pm}$:

$$\hat{J}^2 (\hat{J}_{\pm} |j, m\rangle) = \hbar^2 j(j+1) (\hat{J}_{\pm} |j, m\rangle)$$

$\hat{J}_{\pm}$ operators change the value of m (step up/step down)

$$\begin{aligned} \hat{J}_z (\hat{J}_{\pm} |j, m\rangle) &= ([\hat{J}_z, \hat{J}_{\pm}] + \hat{J}_{\pm} \hat{J}_z) |j, m\rangle \\ &= (\pm \hbar \hat{J}_{\pm} + \hbar m \hat{J}_{\pm}) |j, m\rangle \\ &= \hat{J}_{\pm} \hbar (m \pm 1) |j, m\rangle \\ &= \hbar (m \pm 1) (\hat{J}_{\pm} |j, m\rangle) \end{aligned}$$

Now to Determine this $\leftarrow$

$$\hat{J}_{\pm} |j, m\rangle = \underbrace{C_{\pm}(j, m)}_{\text{constant}} |j, m \pm 1\rangle$$

adjoint $\rightarrow \langle j, m | \hat{J}_{\mp} = \langle j, m \pm 1 | C_{\pm}^*(j, m)$

Overlap $\rightarrow \langle j, m | \hat{J}_{\mp} \hat{J}_{\pm} |j, m\rangle = |C_{\pm}(j, m)|^2$

Recall: $J_{\pm} J_{\mp} = J^2 - J_z^2 \mp \hbar J_z$

$$|C_{\pm}(j,m)|^2 = \langle j,m | (J^2 - J_z^2 \mp \hbar J_z) | j,m \rangle = \hbar^2 j(j+1) - \hbar^2 m^2 \mp \hbar^2 m$$

$$|C_{\pm}(j,m)|^2 = \hbar^2 [j(j+1) - m(m \pm 1)] = \|\hat{J}_{\pm} |j,m\rangle\|^2$$

$$J_{\pm} |j,m\rangle = \hbar \sqrt{j(j+1) - m(m \pm 1)} |j, m \pm 1\rangle$$

$\Rightarrow$ How far can we raise or lower 'm'?

1. $\|\hat{J}_{+} |j,m\rangle\|^2 \geq 0$ [Raising]

$$\hookrightarrow j(j+1) - m(m+1) \geq 0 \Rightarrow -j-1 \leq m \leq j$$

Troubles with Raising:

Let $m = j - \beta$; $0 < \beta < 1$

raise once

$$\hookrightarrow m' = m+1 > j \quad \times$$

$$\hookrightarrow \text{So at } m=j \rightarrow C_{+}(j,j) = 0 \\ \Rightarrow \hat{J}_{+} |j,j\rangle = 0 : \text{Zero state}$$

2. $\|\hat{J}_{-} |j,m\rangle\|^2 \geq 0$ [Lowering]

$$\hookrightarrow -j \leq m \leq j+1$$

Zero state at: $m = -j$

Quantization:

Since m is inc/dec in int, the distance $2j$ b/w j & $-j$ should be an int as well:

$$2j \in \mathbb{Z} \rightarrow j \in \frac{\mathbb{Z}}{2} \rightarrow j = 0, \frac{1}{2}, 1, \frac{3}{2}, 2, \dots$$

Angular momentum can be an Integral or Half Integral.

$\Rightarrow$ For $j = \frac{1}{2}$ we have $|j, m\rangle$ states as:

$$m = j, j-1, \dots, -j$$

$$(j) \quad (-j)$$

$$m = \frac{1}{2}, -\frac{1}{2}$$

$$\left(\frac{1}{2} - 1 = -\frac{1}{2}\right)$$

$$\text{States: } |j, m\rangle = \left|\frac{1}{2}, \frac{1}{2}\right\rangle \text{ \& \& } \left|\frac{1}{2}, -\frac{1}{2}\right\rangle$$
$$\begin{array}{cc} \parallel |+\rangle & \parallel |-\rangle \\ S_z = +\frac{\hbar}{2} & S_z = -\frac{\hbar}{2} \end{array}$$

$\Rightarrow$ Notable results:

$$\frac{1}{\hbar} J \simeq j + \frac{1}{2} + O(1/j)$$

$\uparrow$

$$\frac{1}{\hbar} J = \sqrt{j(j+1)}$$

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The Free Particle and Infinite Sphere Well

For a Free particle: $V(r) = 0$

So

$$\text{Radial Eq: } -\frac{\hbar^2}{2m} \frac{d^2 u_{El}}{dr^2} + \frac{\hbar^2}{2m} \frac{l(l+1)}{r^2} u_{El} = E u_{El}$$

$$\left(-\frac{d^2}{dr^2} u_{El} + \frac{l(l+1)}{r^2} u_{El} \right) = k^2 u_{El}$$

$$\text{where } k = \sqrt{\frac{2mE}{\hbar^2}}$$

$$\text{Let } \rho = kr$$

$$\rightarrow -\frac{d^2 u_{El}}{d\rho^2} + \frac{l(l+1)}{\rho^2} u_{El} = u_{El}$$

$$\downarrow$$
$$\text{Soln: } u_{El} = c \cdot \tilde{j}_l(\rho)$$

$\tilde{j}_l \rightarrow$ Spherical Bessel Functions.

$$u_{El} = \gamma \tilde{j}_l(kr)$$

For a Free particle:

$$\psi_{Elm}(r) = \tilde{j}_l(kr) \cdot Y_{lm}(\theta, \phi)$$

$$\bullet \quad r \rightarrow \infty : u_{El} \sim \sin\left(kr - \frac{l\pi}{2}\right)$$

↳ Whenever the potential is not zero, but vanishes beyond some radius

$$\hookrightarrow U_{El} \sim \sin\left(kr - \frac{l\pi}{2} + \delta_l(E)\right)$$

$\delta_{El} \rightarrow$ Phase shift = 0 if no Potential

$\Rightarrow$ Infinite Sphere Well:

Forces a particle to be in $r=a$ sphere

$$V(r) = \begin{cases} 0 & r \leq a \\ \infty & r > a \end{cases}$$

Radial eq: $-\frac{d^2 U_{El}}{dr^2} + \frac{l(l+1)}{r^2} U_{El} = U_{El} \quad ; \quad p = kr$
 $\quad \quad \quad ; \quad k = \sqrt{\frac{2mE}{\hbar^2}}$

$\Rightarrow$ For $l=0$:

$$-\frac{d^2 U_{E0}}{dr^2} = U_{E0} \rightarrow U_{E0} = A \sin p + B \cos p$$

with B.C $\rightarrow$ $U_{E0}(r) = \sin kr$

with $ka = n\pi$

$\hookrightarrow$ So, $E_{n,0} = \underbrace{\frac{\hbar^2}{2ma^2}}_{\text{natural energy scale}} n^2 \pi^2$: Energy is quantized.

Unit Free Energy: $E_{n,l} = \frac{2ma^2}{\hbar^2} E_{n,l}$

For $l=0$

$$\hookrightarrow E_{n,0} = n^2 \pi^2$$

$\Rightarrow$ For $l=1$:

$$u_{E,1} = r j_1(kr)$$

$$j_1(p) = \frac{\sin p}{p^2} - \frac{\cos p}{p}$$

Let $z_{n,l}$ denote zero of j_l : $j_l(z_{n,l}) = 0$
So,

$$k_{n,l} a = z_{n,l}$$

$$\hookrightarrow \boxed{E_{n,l} = z_{n,l}^2}$$

The 3D Isotropic Oscillator

$$V = \frac{1}{2} m \omega^2 (x^2 + y^2 + z^2) = \frac{1}{2} m \omega^2 r^2$$

This quantum 3D oscillator is a lot more symmetric than infinite sphere well

Using creation / annihilation operators: ($\hat{N} = \hat{a}^\dagger \hat{a}$)

$$H = \hbar \omega \left(\hat{N}_1 + \hat{N}_2 + \hat{N}_3 + \frac{3}{2} \right) = \hbar \omega \left(\hat{N} + \frac{3}{2} \right)$$

We describe the tensor product in 3D as:

$$H_{(3D)} = H_x \otimes H_y \otimes H_z$$

Basis states: $H_x : (\hat{a}_x^\dagger)^{n_x} |0\rangle_x ; n_x = 0, 1, \dots$

$$H_y : (\hat{a}_y^\dagger)^{n_y} |0\rangle_y$$

$$H_z : (\hat{a}_z^\dagger)^{n_z} |0\rangle_z$$

★ For H_{3D} will be just tensor product of these

simplified: H_{3D} basis $\rightarrow (\hat{a}_x^\dagger)^{n_x} (\hat{a}_y^\dagger)^{n_y} (\hat{a}_z^\dagger)^{n_z} |0\rangle$

Thus:

① $N=0$:

$$E = \frac{3}{2} \hbar \omega$$

State: $|0\rangle \leftrightarrow l=0$

② $N=1$:

$$E = \frac{5}{2} \hbar \omega$$

States: $\hat{a}_x^\dagger |0\rangle, \hat{a}_y^\dagger |0\rangle, \hat{a}_z^\dagger |0\rangle$

no. of states.



l	#
0	1
1	3
2	5
3	7
4	9
5	11
6	13
7	15

② $N=2$:

$$E = \frac{7}{2} \hbar \omega$$

States: $(\hat{a}_x^\dagger)^2 |0\rangle, (\hat{a}_y^\dagger)^2 |0\rangle, (\hat{a}_z^\dagger)^2 |0\rangle,$
 $(\hat{a}_x^\dagger)(\hat{a}_y^\dagger) |0\rangle, (\hat{a}_y^\dagger)(\hat{a}_z^\dagger) |0\rangle, (\hat{a}_z^\dagger)(\hat{a}_x^\dagger) |0\rangle$

Say to have 6 states:

we get 5 from $l=2$ & 1 from $l=0$

So,

$$\text{Sum } N=2 \text{ states: } (l=2) \oplus (l=0)$$

"Correct Sum"

Thus,

$$\text{No. of states: } \binom{N+d-1}{d-1}$$

d : no. of spatial dim

③ $N=3$:

$$E = \frac{9}{2} \hbar \omega$$

$$\text{States: } \binom{3+3-1}{2} = 10$$

$$\text{Possibilities: } (l=3) \oplus (l=1) \text{ or } (l=4) \oplus (l=0)$$

not possible

$$\Rightarrow (l=3) \oplus (l=1)$$

④ $N=4$:

$$E = \frac{11}{2} \hbar \omega$$

$$\text{States: } \binom{4+3-1}{2} = 15$$

$$\Rightarrow (l=4) \oplus (l=2) \oplus (l=0) \quad \left| \begin{array}{l} l=7 \text{ not possible} \\ \text{as } 4, 5, 6 \text{ have not} \\ \text{appeared yet.} \end{array} \right.$$

General Solution:

$$\text{States are in the form of: } (a_x^\dagger)^{n_x} (a_y^\dagger)^{n_y} (a_z^\dagger)^{n_z} |0\rangle$$

where,

$$n_x + n_y + n_z = N$$

$$\# N = 1 + 2 + \dots + (N+1) \\ = \frac{(N+1)(N+2)}{2}$$



$\Rightarrow \hat{a}_L$ and $\hat{a}_R$ operators :

$$\hat{a}_L = \frac{1}{\sqrt{2}} (\hat{a}_x + i\hat{a}_y) \quad ; \quad \hat{a}_R = \frac{1}{\sqrt{2}} (\hat{a}_x - i\hat{a}_y)$$

$$\bullet [\hat{a}_L, \hat{a}_L^\dagger] = [\hat{a}_R, \hat{a}_R^\dagger] = 1$$

$$\bullet \hat{N}_R = \hat{a}_R^\dagger \hat{a}_R \quad ; \quad \hat{N}_L = \hat{a}_L^\dagger \hat{a}_L$$

$$\Rightarrow H = \hbar\omega \left(\hat{N}_R + \hat{N}_L + \hat{N}_z + \frac{3}{2} \right)$$

$$\Rightarrow \hat{L}_z = \hbar (\hat{N}_R - \hat{N}_L)$$

★ Thus the new states are:

• $N=1$

$$a_R^\dagger |0\rangle, a_z^\dagger |0\rangle, a_L^\dagger |0\rangle$$

$$L_z = \hbar \quad L_z = 0 \quad L_z = -\hbar$$

So for an value N , the highest L_z state is $(a_R^\dagger)^N |0\rangle$
 $= \hbar N$

Thus we should have $l=N$ multiplet

Consider:

$$|a\rangle, |b\rangle, \dots, (a_R^\dagger)^N |0\rangle$$

$$\begin{aligned} L_z |h\rangle &= N \\ \hat{L}_z |h\rangle &= N-1 \\ \hat{L}_z |h\rangle &= N-2 \end{aligned}$$

$$\begin{aligned} (a_R^\dagger)^{N-1} a_L^\dagger |0\rangle \\ (a_R^\dagger)^{N-2} (a_L^\dagger)^2 |0\rangle \\ \text{or} \\ (a_R^\dagger)^{N-1} a_L^\dagger |0\rangle \end{aligned} \left. \vphantom{\begin{aligned} (a_R^\dagger)^{N-1} a_L^\dagger |0\rangle \\ (a_R^\dagger)^{N-2} (a_L^\dagger)^2 |0\rangle \\ (a_R^\dagger)^{N-1} a_L^\dagger |0\rangle \end{aligned}} \right\} \underline{2 \text{ states}}$$

Todo: You need to do this again!!

